30 Uranium-235 undergoes fission in a reaction shown below, releasing 195 MeV of energy. The binding energy per nucleon for uranium-235 is 7.6 MeV. The binding energy per nucleon for caesium and rubidium are each approximately equal to **X** MeV.

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{143}_{55}Cs + {}^{90}_{37}Rb + 3\,{}^{1}_{0}n$$

What is the value of **X**?

A 6.7 **B** 6.8 **C** 8.4 **D** 8.5

Ans: **D**

$$\text{Energy released} = BE_{products} - BE_{reactants}$$

$$195 = X(143) + X(90) - 7.6(235)$$

$$X = 8.50 \text{ MeV}$$